Stochastic Processes Semester 1

Tutorial 5 – Solutions

1. (a) There are three closed classes: $C_1 = \{6\}$, $C_2 = \{1, 4\}$, and $C_3 = \{3, 5\}$ and one open class $O_1 = \{2\}$. Here
   - C_1 is closed because it is absorbing: $p_{66} = 1$;
   - C_2 is closed because $p_{14} > 0$,$p_{41} > 0$, and $p_{11} + p_{14} = 1$, $p_{41} + p_{44} = 1$;
   - C_3 is closed because $p_{35} > 0, p_{53} > 0$, and $p_{33} + p_{35} = 1$, $p_{53} + p_{55} = 1$;
   - O_1 is open because, say, $p_{21} > 0$, but $p_{i2} = 0$ for all $i \in C_2$ (i.e. $p_{12} = p_{42} = 0$).

   (b) State 3 belongs to the finite closed class C_3 which is aperiodic (since, say, $p_{33} > 0$). Therefore there exists a unique stationary distribution in this class, and the limiting distribution for any initial value within this class coincides with the stationary one. To find the stationary distribution, we solve the system of equations:

$$\pi_3^{(3)} = \pi_3^{(3)} \cdot \frac{1}{4} + \pi_5^{(3)} \cdot \frac{1}{4};$$
$$\pi_3^{(3)} + \pi_5^{(3)} = 1$$

   which has the unique solution $\pi_3^{(3)} = 1/4$; $\pi_5^{(3)} = 3/4$. Thus, the limiting distribution for the initial state $X_0 = 3$ is $\boldsymbol{\pi}^{(3)} = (0, 0, 1/4, 0, 3/4, 0)$.

   (c) Denote by $\alpha_2^{(i)}$ the probability of absorption in class $i = 1, 2, 3$ for a Markov chain that starts from $X_0 = 2$. We have to find $\alpha_2^{(1)}$. From the total probability formula,

$$\alpha_2^{(1)} = p_{21}\alpha_1^{(1)} + p_{22}\alpha_2^{(1)} + p_{23}\alpha_3^{(1)} + p_{26}\alpha_6^{(1)}$$

   where $\alpha_1^{(1)} = \alpha_3^{(1)} = 0$ and $\alpha_6^{(1)} = 1$. Therefore we get

$$\alpha_2^{(1)} = \frac{1}{8}\alpha_2^{(1)} + \frac{1}{4},$$

   and $\alpha_2^{(1)} = 2/7$.

(d) Here we have to find $\alpha_2^{(2)}$.

From the total probability formula,

$$\alpha_2^{(2)} = p_{21}\alpha_1^{(2)} + p_{22}\alpha_2^{(2)} + p_{23}\alpha_3^{(2)} + p_{26}\alpha_6^{(2)}$$

where $\alpha_1^{(2)} = 1$ and $\alpha_3^{(2)} = \alpha_6^{(2)} = 0$. Therefore we obtain

$$\alpha_2^{(2)} = \frac{3}{8} + \frac{1}{8}\alpha_2^{(2)}$$

and $\alpha_2^{(2)} = 3/7$.

(e) Since the state space is finite,

$$\alpha_2^{(1)} + \alpha_2^{(2)} + \alpha_2^{(3)} = 1.$$

Therefore $\alpha_2^{(3)} = 1 - \alpha_2^{(1)} - \alpha_2^{(2)} = 2/7$.

The closed classes C_1 and C_2 are also aperiodic (C_1 is absorbing, $p_{66} = 1 > 0$ and C_2 is aperiodic since, say, $p_{11} > 0$). Therefore there exists a unique stationary distribution within each of these two classes: $\boldsymbol{\pi}^{(1)} = (0, 0, 0, 0, 0, 1)$ and $\boldsymbol{\pi}^{(2)} = (\pi_1^{(2)}, 0, 0, \pi_4^{(2)}, 0, 0)$. Then the limiting distribution for the initial state $X_0 = 2$ is

$$\alpha_2^{(1)}\boldsymbol{\pi}^{(1)} + \alpha_2^{(2)}\boldsymbol{\pi}^{(2)} + \alpha_2^{(3)}\boldsymbol{\pi}^{(3)}.$$

The only term to find is the vector $\boldsymbol{\pi}^{(2)}$. For that, we solve a system of equations;

$$\begin{aligned} \pi_1^{(2)} &= \pi_1^{(2)} \cdot \frac{1}{3} + \pi_4^{(2)} \cdot \frac{1}{2}; \\ \pi_1^{(2)} + \pi_4^{(2)} &= 1 \end{aligned}$$

which has the solution $\pi_1 = 3/7$, $\pi_4 = 4/7$. Therefore $\boldsymbol{\pi}^{(2)} = (3/7, 0, 0, 4/7, 0, 0)$. Then the limiting distribution for the initial state $X_0 = 2$ is:

$$\left(\frac{9}{49}, 0, \frac{1}{14}, \frac{12}{49}, \frac{3}{14}, \frac{2}{7}\right).$$

2. (a) This is an irreducible Markov chain which is periodic, with period $d = 3$.

Indeed, let $S_1 = \{1, 5\}$, $S_2 = \{2, 4\}$, and $S_3 = \{3, 6\}$. Then $X_0 \in S_1$ if and only if $X_2 \in S_2$ (since $p_{12} + p_{14} = p_{52} + p_{54} = 1$ and $p_{i2} = p_{i4} = 0$ for any $i \neq 1, 5$). Similarly $X_0 \in S_2$ if and only if $X_1 \in S_3$ (since $p_{23} + p_{26} = p_{43} + p_{46} = 1$ and $p_{i3} = p_{i6} = 0$ for any $i \neq 2, 4$) and $X_0 \in S_3$ if and only if $X_1 \in S_1$ (here $p_{31} + p_{35} = p_{61} + p_{65} = 1$ and $p_{i1} = p_{i5} = 0$ for any $i \neq 3, 6$). Thus, the period is 3.

Further, the Markov chain is irreducible since $i \to j$, for any i, j. For example we may show that $1 \to i$, for any i. Indeed, $p_{12} > 0, p_{14} > 0$, so $1 \to 2$ and $1 \to 4$. Since $2 \to 3$ and $2 \to 6$, we get that also $1 \to 3$ and $1 \to 6$. Finally, $6 \to 5$ implies that $1 \to 5$.

(b) The answers are: no, no, no, yes, no, and yes. The reasoning is: a limiting distribution exists if and only if the initial distribution μ is such that

$$\mu_1 + \mu_5 = \mu_2 + \mu_4 = \mu_3 + \mu_6 = \frac{1}{3}.$$

3. (a) Here is the solution using the reduction argument. It is clear that this Markov chain contains a single closed class and is periodic with period $d = 2$. Indeed, $p_{12}+p_{14} = p_{32}+p_{34} = 1$ and $p_{21}+p_{23} = p_{41} + p_{43} = 1$ and, say, $p_{11}^{(2)} \geq p_{12}p_{21} > 0$ and, similarly, $p_{11}^{(2k)} > 0$ for any $k = 2, 3, \ldots$.

Consider the two-step transition matrix ("two-step" because $d = 2$):

$$P^2 = \begin{pmatrix} 4/9 & 0 & 5/9 & 0 \\ 0 & 11/18 & 0 & 7/18 \\ 3/8 & 0 & 5/8 & 0 \\ 0 & 13/24 & 0 & 11/24 \end{pmatrix}$$

The Markov chain with 1-step transition matrix $Q = P^2$ has two closed classes, $\{1, 3\}$ and $\{2, 4\}$.

Find the unique stationary distribution in each class, by solving

equations:

$$\begin{aligned} \frac{4}{9}\pi_1 + \frac{3}{8}\pi_3 &= \pi_1, \\ \frac{5}{9}\pi_1 + \frac{5}{8}\pi_3 &= \pi_3, \\ \pi_1 + \pi_3 &= 1, \end{aligned}$$

and

$$\begin{aligned} \frac{11}{18}\pi_2 + \frac{13}{24}\pi_4 &= \pi_2, \\ \frac{7}{18}\pi_2 + \frac{11}{24}\pi_4 &= \pi_4, \\ \pi_2 + \pi_4 &= 1. \end{aligned}$$

Then divide each probability π_i by $d = 2$ to obtain the stationary distribution of the original Markov chain: $(\pi_1/2, \pi_2/2, \pi_3/2, \pi_4/2)$.

The first equation implies $(5/9)\pi_1 = (3/8)\pi_3$, then $\pi_3 = (40/27)\pi_1$. Substituting this into $\pi_1 + \pi_3 = 1$ gives $\pi_1 + (40/27)\pi_1 = 1$, then $\pi_1 = 27/67$ and $\pi_3 = 40/67$.
In the second set of equations, the first equation implies $(13/24)\pi_4 = (7/18)\pi_2$, then $\pi_4 = (28/39)\pi_2$. Substituting this into $\pi_2 + \pi_4 = 1$ gives $\pi_2 + (28/39)\pi_2 = 1$, then $\pi_2 = 39/67$ and $\pi_4 = 28/67$.
Dividing all of the above by the period $d = 2$ gives the stationary distribution for the original Markov chain as

$$(27/134, 39/134, 20/67, 14/67).$$

[You may check that these 4 numbers sum up to 1].

(b) A limiting distribution exists iff the initial distribution $\boldsymbol{\mu} = (\mu_1, \mu_2, \mu_3, \mu_4)$ is such that $\mu_1 + \mu_3 = 1/2$ and $\mu_2 + \mu_4 = 1/2$.

4. We already know that there are two closed classes here, $C_1 = \{1, 3, 6\}$ and $C_2 = \{4\}$, and that the absorption probability into the class $\{4\}$ for a Markov chain starting from 2 is $\alpha_2^{\{4\}} = 1/8$. Then the probability to be absorbed in the class $\{1, 3, 6\}$ is

$$\alpha_2^{\{1,3,6\}} = 1 - \alpha_2^{\{4\}} = 7/8.$$

Within each closed class, we may find the unique stationary distribution in that class. Clearly, within the class $\{4\}$, the stationary distribution is $\pi_4^{(2)} = 1$, i.e. $\pi^{(\mathbf{2})} = (0, 0, 0, 1, 0, 0, 0)$. For the class $\{1, 3, 6\}$, we have to solve the system of equations

$$\begin{aligned} \frac{1}{2}\pi_3 + \frac{1}{2}\pi_6 &= \pi_1 \\ \frac{1}{2}\pi_1 + \frac{1}{2}\pi_3 &= \pi_3, \\ \frac{1}{2}\pi_1 + \frac{1}{2}\pi_6 &= \pi_6, \\ \pi_1 + \pi_3 + \pi_6 &= 1. \end{aligned}$$

The second equation tells us that $\pi_1 = \pi_3$ while the third equation tells us that $\pi_1 = \pi_6$. It follows that $\pi_1 = \pi_3 = \pi_6 = 1/3$ since the sum of these 3 identical numbers is 1. Then $\pi^{(\mathbf{1})} = (1/3, 0, 1/3, 0, 0, 1/3, 0)$.

Finally, the limiting distribution is:

$$(\alpha_2^{\{1,3,6\}}\pi_1, 0, \alpha_2^{\{1,3,6\}}\pi_3, \alpha_2^{\{4\}}\pi_4, 0, \alpha_2^{\{1,3,6\}}\pi_6, 0) = (7/24, 0, 7/24, 1/8, 0, 7/24, 0).$$